Binomial Hypothesis testing finding p value

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Question: how to find a p-value where:

$H_0 : \hat{p} = 1/6$
$H_a : \hat{p} \neq 1/6$
Significance level is $\alpha = .05$.

In my test: $n = 60$, $x = 14$, so my $\hat{p} = 0.2333$.

How do I find a p-value using a two-tailed binomial distribution?

probability hypothesis-testing distributions binomial-distribution

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edited Mar 9, 2021 at 14:46
Dave
65k 7 101 286

asked Mar 9, 2021 at 14:38
Raymond Li
1

1 I see two errors in your question. First, you know exactly what $\hat{p}$ is. The test is of p itself. Second, the p-value does not depend on α. // I cleaned up your mathematical notation. Please check that I did not change any of the meaning. – Dave Mar 9, 2021 at 14:49

yep thx u didnt mess the question. – Raymond Li Mar 9, 2021 at 15:10

1 The equations "$\hat{p} = 1/6$" and "$\hat{p} \neq 1/6$" do not articulate hypotheses, *because they are assertions about what you observed* and not about the distribution. Most likely you want to drop the hats. – whuber ♦ Mar 9, 2021 at 15:40

1 Answer

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1

It seems you want to test $H_0 : p = 1/6$ against $H_a : p \neq 1/6$. You have $n = 60$ Bernoulli trials with $X = 14$ successes. Assuming H_0 is true $X \sim \text{Binom}(n = 60, p = 1/6)$. You want to reject H_0 for values of X smaller the 10 and larger than 10.

Exact binomial test. Specifically, if you reject for $\{X \le 4\}$ or $\{X \ge 17\}$, then you will have a test at level $\alpha = 0.0366$. which as large a significance level as I can find without going over $0.05 = 5\%$. [The binomial distribution is discrete, so (nonrandomized) tests at *exactly* the 5% level are usually not possible.]

Therefore, with observed $X = 14$, you do not have evidence to reject H_0. [Computations use R software.]

```
sum(dbinom(c(0:4,17:60), 60, 1/6))
[1] 0.03663785
```

The **P-value** of the test is the probability of a more extreme result than than observed [in the direction(s) of the alternative]. For your test $P(X \ge 14) = 0.0648$. Because this is a two-sided test, one convention is to double that probability to get P-value 0.1296. One rejects H_0 at the 5% level when the P-value is less than 5%.

```
1 - pbinom(14, 60, 1/6)
[1] 0.06478038
2*(1 - pbinom(14, 60, 1/6))
[1] 0.1295608
```

***Note:* Approximate normal test.** Because you have $n = 60$ trials you can use an approximate normal test, using the statistic

$$z = \frac{X - np}{\sqrt{np(1-p)}} = \frac{14 - 10}{1.385641} = 1.386.$$

Then you would not reject (approximately) at the 5% level because z lies between ± 1.96. [Using the continuous normal distribution may make it seem that a test at exactly the 5% level is possible, but the z statistic still takes discrete fractional values. So z values exactly equal to ± 1.96 would not generally be possible.]

The **P-value** of the approximate normal test is $2P(Z > x) = 2P(Z > 1.386) = 0.166 > 0.05$, where Z is a standard normal random variable.

```
1 - pnorm(1.386)
[1] 0.08287345
2*(1 - pnorm(1.386))
[1] 0.1657469
```

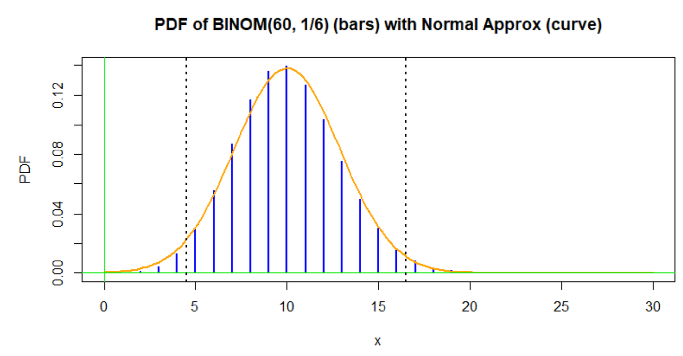
Here is output from the z test using Minitab statistical software (with the normal approximation):

```
Test and CI for One Proportion

Test of p = 0.166667 vs p ≠ 0.166667

Sample    X   N  Sample p         95% CI          Z-Value  P-Value
1        14  60  0.233333  (0.126314, 0.340353)     1.39    0.166
```

The figure below shows the null binomial distribution with the density function of its normal approximation (from matching means and standard deviations between binomial and normal distributions). The rejection region is outside the two vertical dotted lines.



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56.9k 2 34 94